



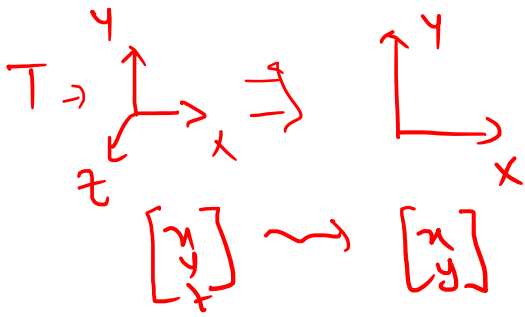
→ Linear Transformations

⇒ Change of Basis

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Recap

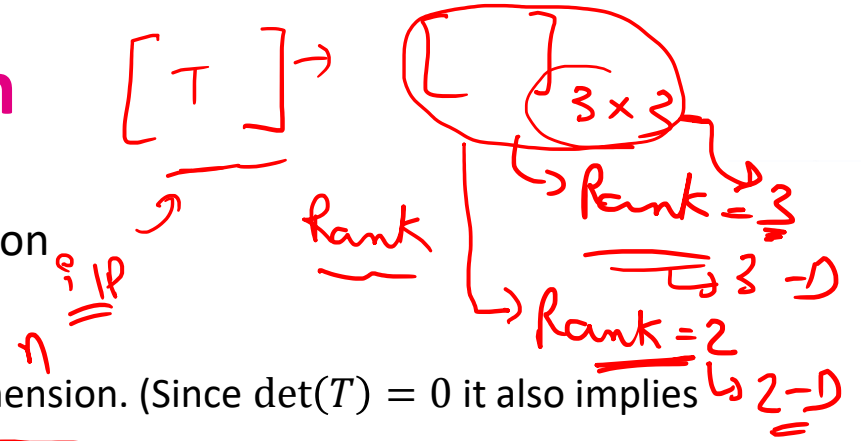
- Matrices →
- Vectors →
- Basis and unit vectors
- Linear combination of vectors
- Linear Transformations



Linear Transformation

Some properties of Linear Transformation

$$\det(T) = 0$$



- When $\det(T) = 0$ the transformation T reduces the given vector to lower dimension. (Since $\det(T) = 0$ it also implies inverse of the given matrix doesn't exist)
 - For a vector in 2-D space this type of transformation can result into a line or a point.
 - Similarly, for a vector in 3-D space this type of transformation can result into a vector space in 2-D space (plane) or a line or a point.
 - This is determined with the rank of the transformation matrix T
- A full rank transformation matrix does not reduce the dimension of the transformed vector space.
- Column space: the set of all possible outputs for $A\vec{v}$ is the **column space** of transformation A . (in simpler words it is the span of the columns of your transformation matrix)

$A: \vec{v} \rightarrow \vec{w}$ → Column Space
- For linear transformation the vector 0 is always in the column space
- For transformations which result in lower dimension of the transformed vector space will have multiple vectors that will be mapped to 0
 - These set of vectors which are mapped to origin is commonly as **null space**. It is the set of vectors which becomes null after the transformation

$\vec{v} \rightarrow 0$

Non-Square Matrices

There can be a transformation which transforms a 2-D vector to a 3-D vector

$$\rightarrow \underline{\underline{3D}} \rightarrow \underline{\underline{2D}}$$

$$\underline{\underline{4D}} \rightarrow \underline{\underline{3D}} / \underline{\underline{2D}}$$

$$\rightarrow \begin{bmatrix} x \\ y \end{bmatrix} \xrightarrow{T} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$T = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \quad \begin{matrix} \text{rows} \\ \text{columns} \end{matrix} \quad \underline{\underline{2D}} \rightarrow \underline{\underline{3D}}$$

non-square matrix

$$\underline{\underline{3D}} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \xrightarrow{T} \underline{\underline{2D}} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$T = \begin{bmatrix} a & c & e \\ b & d & f \end{bmatrix}_{2 \times 3} \rightarrow \begin{matrix} \underline{\underline{3D}} \text{ i/p} \\ \underline{\underline{2D}} \text{ o/p} \end{matrix}$$

non-full rank

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}_{3 \times 2} \rightarrow \text{Rank}=2$$

Change of Basis

In many real-world applications, a problem describes using a given coordinate system may be solved more easily by switching to a new coordinate system.

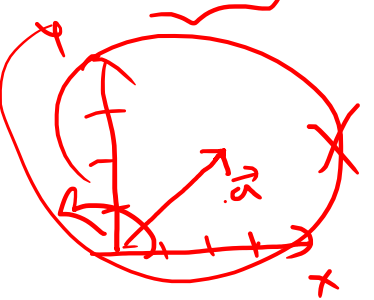
This switching to a new coordinate system is also sometimes known as **change of basis**.

Translation

existing (ref.) $\rightarrow$ (new)

E.X.

$$\vec{a}_{ref} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3\hat{i} + 2\hat{j}$$



$$\vec{a}_{new} = \begin{bmatrix} 5 \\ 3 \\ 1 \\ 3 \end{bmatrix}$$

unit vectors $\Rightarrow$ $\vec{b}_1, \vec{b}_2$
 $\hat{i}, \hat{j}$

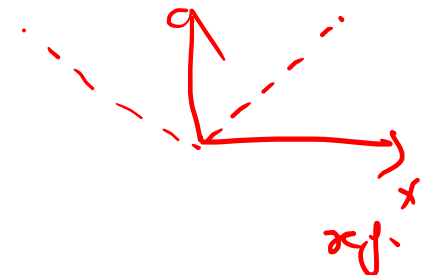
$$\vec{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\vec{b}_1 = 2\hat{i} + \hat{j}, \vec{b}_2 = -\hat{i} + \hat{j}$$

$\vec{a}_{new}$
 $\vec{a}_{ref}$

$\rightarrow$ Not

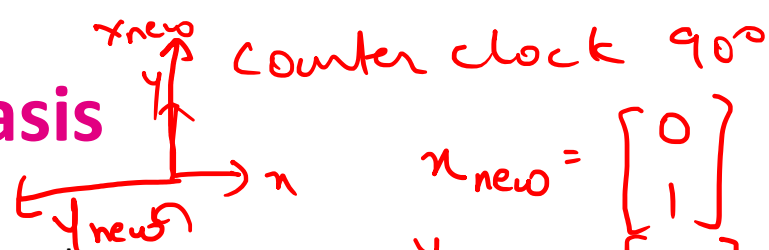
The origin should be the same



$$\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix} = x\hat{i} + y\hat{j}$$

Change of Basis

How to translate between various coordinate systems



$$x_{\text{new}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad y_{\text{new}} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Special Case

$$\vec{a}_{\text{new}} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \Rightarrow = -1\vec{b}_1 + 2\vec{b}_2 = -\vec{b}_1 + 2\vec{b}_2$$

$$\vec{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \vec{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\vec{a}_{\text{ref}} = -1 \underbrace{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}_{\vec{b}_1} + 2 \underbrace{\begin{bmatrix} -1 \\ 1 \end{bmatrix}}_{\vec{b}_2} \quad (\hat{i}, \hat{j})$$

$$= \begin{bmatrix} -2 \\ -1 \end{bmatrix} + \begin{bmatrix} -2 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$$

$$\vec{a}_{\text{ref}} = \begin{bmatrix} -4 \\ 1 \end{bmatrix} = -4\hat{i} + \hat{j} \Rightarrow \text{ref.} \\ \approx -\vec{b}_1 + 2\vec{b}_2 \Rightarrow \text{new}$$

Linear transformation $\rightarrow$ Special Case

$$[\vec{b}_1, \vec{b}_2] = T$$

$$T \cdot \vec{a}_{\text{new}} = \vec{a}_{\text{ref}}$$

$$\vec{a}_{\text{ref}} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$T = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \rightarrow T^{-1} = \begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 2/3 \end{bmatrix}, \quad \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 5/3 \\ 1/3 \end{bmatrix}$$

$$\vec{a}_{\text{new}}$$

$$\vec{a}_{\text{new}} = T^{-1} \cdot \vec{a}_{\text{ref}}$$

$$T^{-1} \cdot \vec{a}_{\text{ref}} = \vec{a}_{\text{new}}$$

Change of Basis

$$A \cdot \vec{a}_{\text{new}} = \vec{a}_{\text{ref.}}$$

Similarly

$$\vec{a}_{\text{new}} = A^{-1} \vec{a}_{\text{ref.}}$$

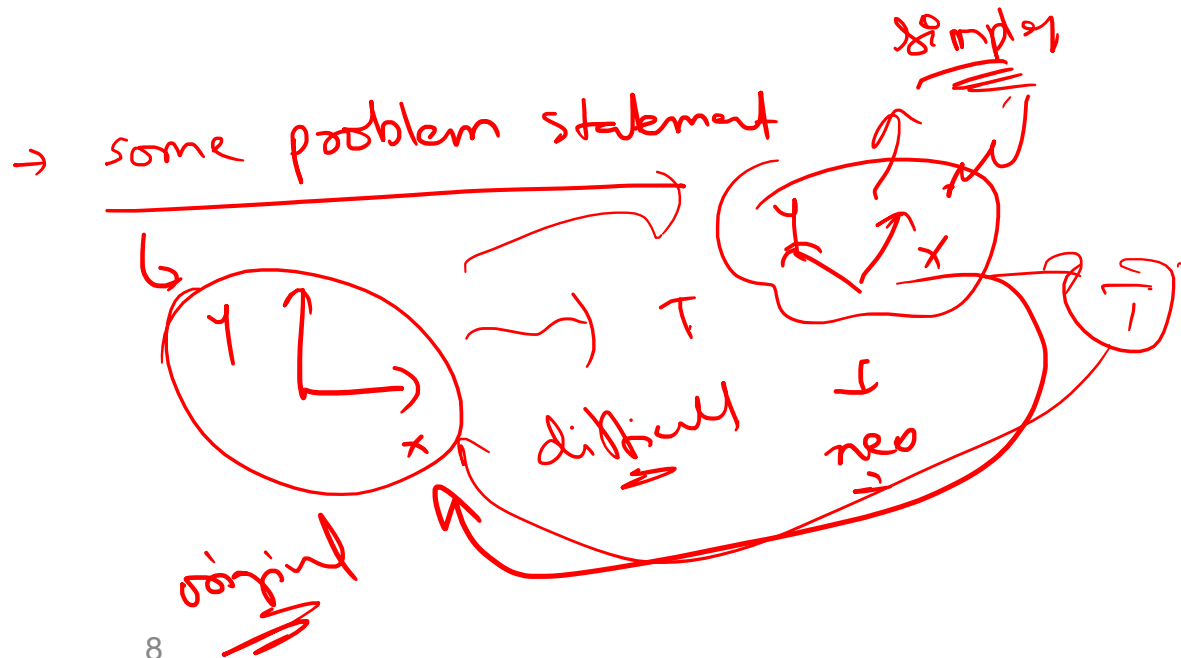
$$A^{-1} \cdot T \cdot A \cdot \vec{V}$$

→ composition → series of transformations

Change of Basis

↳ operation

↳ matrices





Thank you!